# Prototype Detection Samples – Faster R-CNN on Full-Body X-ray Scans

**Green boxes = Ground truth annotations**

**White boxes = Model predictions**

**Each box label format: Class name [confidence score]**

**Class mapping (selected examples):**

**1 = surgical_implant**

**2 = shoe**

**3 = dental**

**4 = zipper**

**5 = zipper_buttons**

**Numbers on white boxes = confidence scores .**

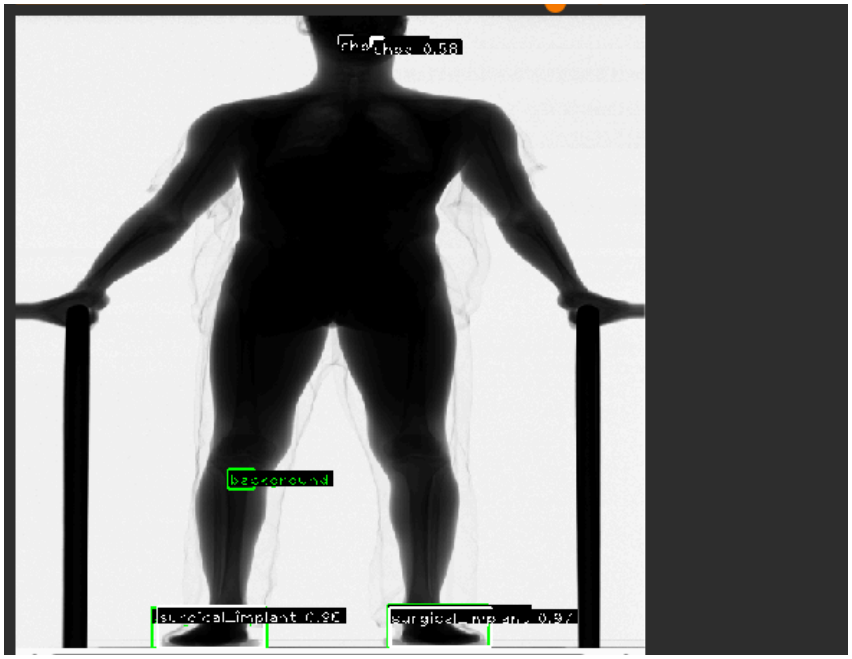


**Figure 1: Raw X-ray image with both ground truth (green) and predictions (white) – wrong detection of multiple classes**

This image shows a full-body scan with ground truth (green) and model predictions (white) overlaid.

- Green boxes: Annotated ground truth (e.g., shoe , zipper ).
- White boxes: Model detections (e.g., surgical_implant  with high confidence).  major mismatches visible → Due to background counted as a class.

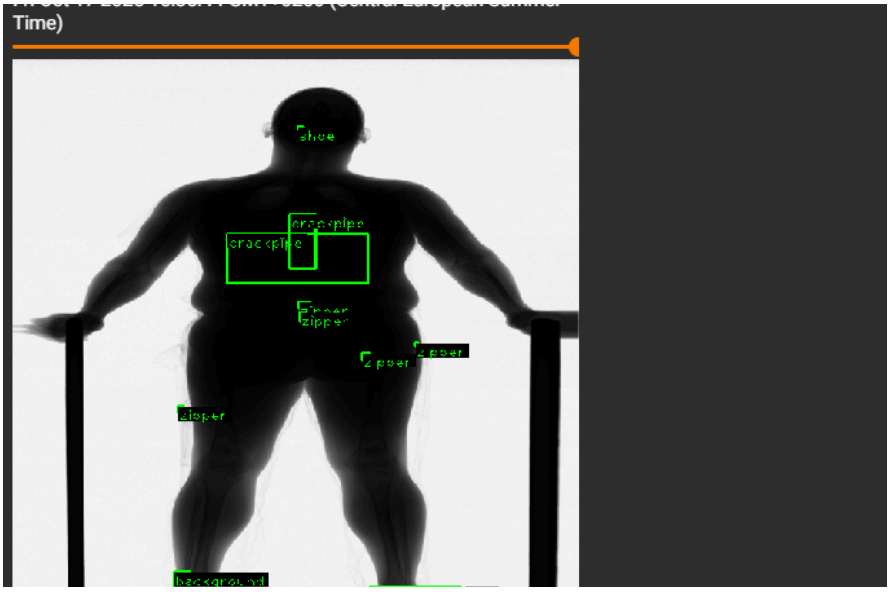


**Figure 2: Challenge – No predictions despite ground truth annotations**

This raw image shows several ground truth objects (green boxes), but **no white prediction boxes** were generated by the model.

- Visible ground truth: e.g., zipper_buttons [5], possibly dental [3] or small contraband.
- Issue: Model failed to propose regions → low recall on small/low-contrast items (recall 0.395 overall with $\log_2$, see Table 1). This highlights the challenge of class imbalance and small objects occupying <1% of image area.

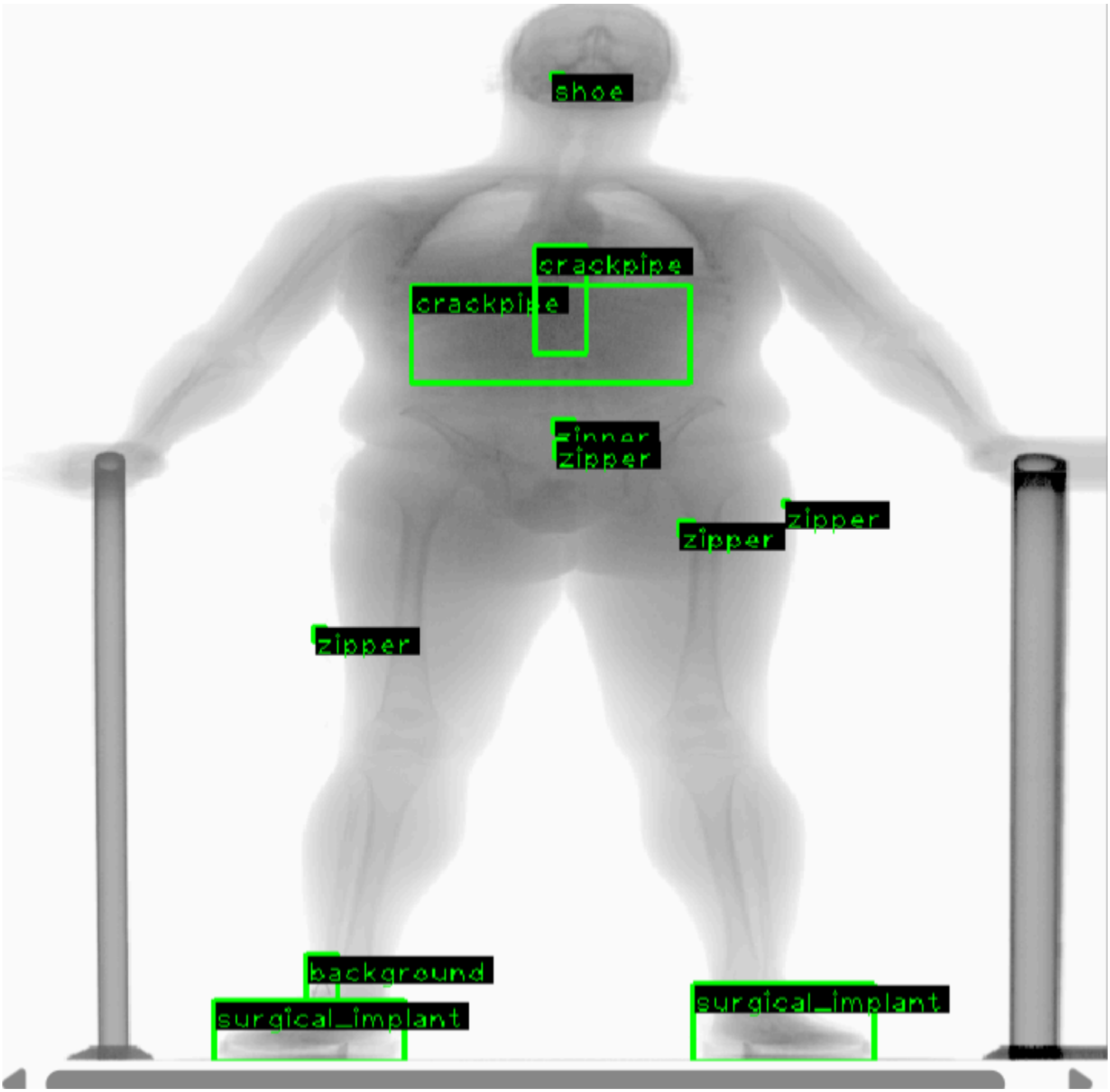


**Figure 3: An Image with Log 2 Filter in preprocess, but with bounding boxes with mismatched classes due to the background being counted as a class.**

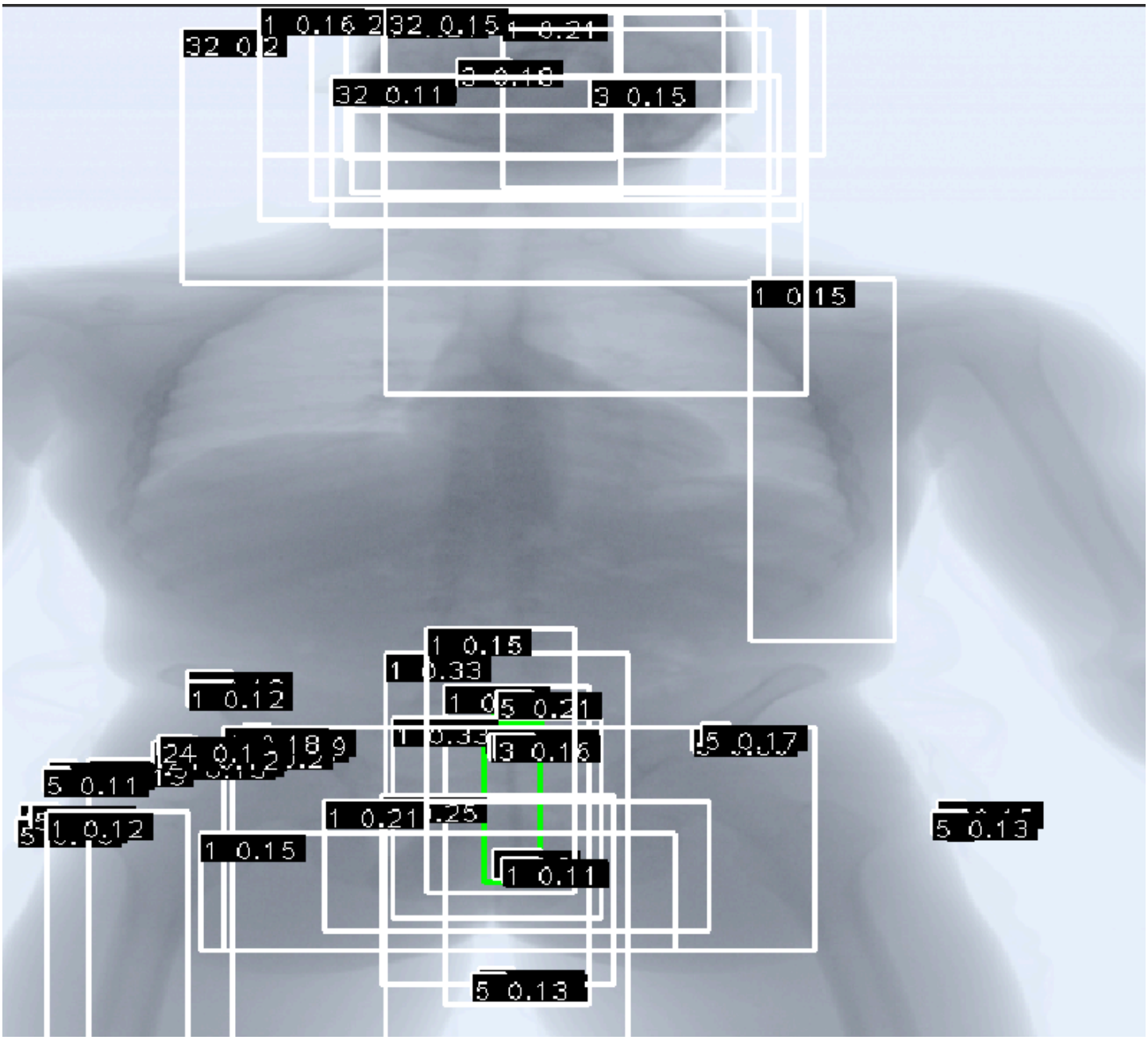


**Figure 4: Overfitting and overlapping predictions without NMS**

Image showing multiple overlapping white prediction boxes on the same object (likely zipper_buttons [5] ).

- low confidence scores on overlapping regions .
- Issue: Model over-predicts due to lack of Non-Maximum Suppression (NMS) → high false positives (see FP/TP ratios in Experiment_Results.pdf). Fixed in later runs by adding NMS and lowering confidence threshold to 0.3 (November/December meeting notes).

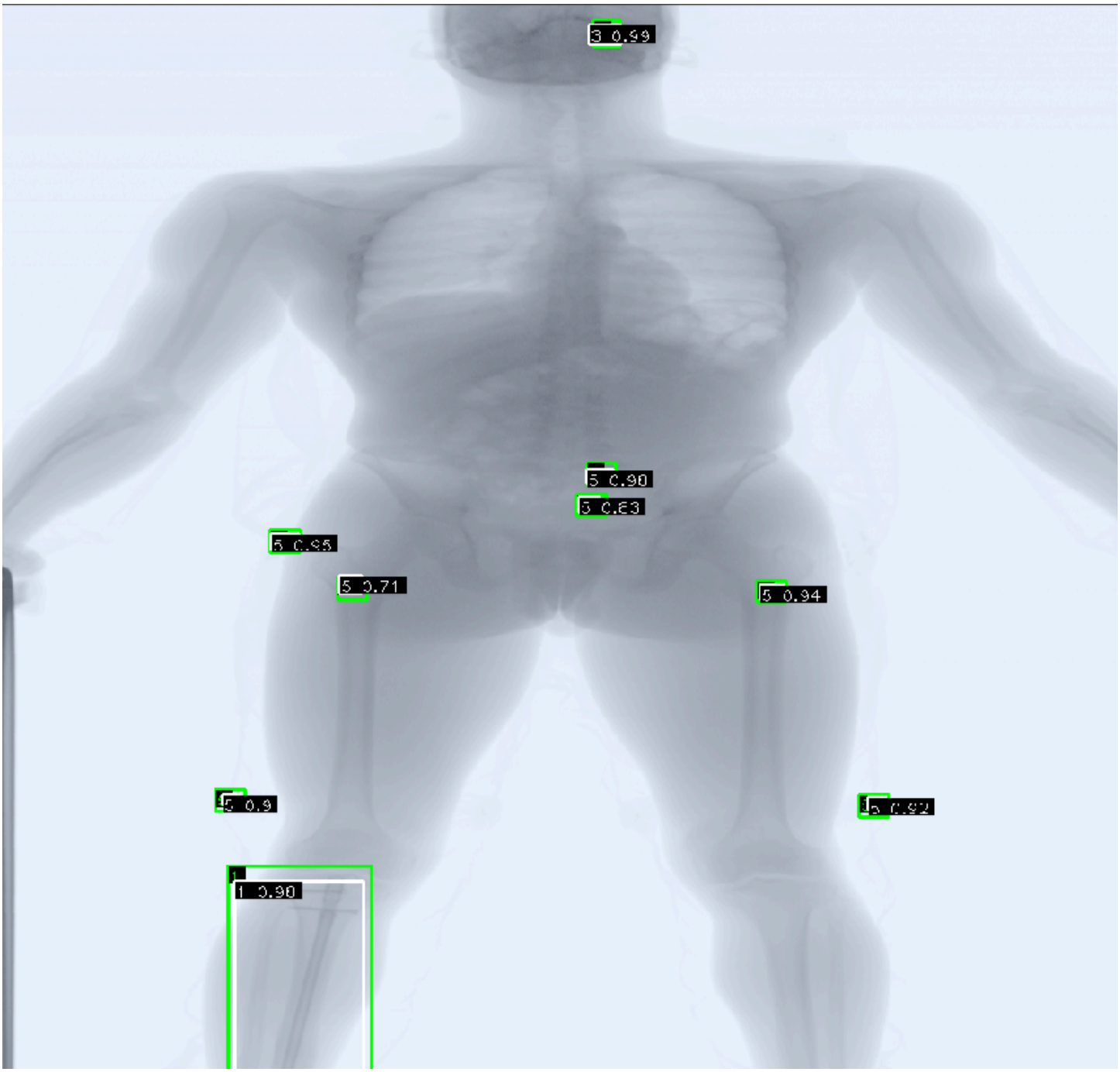


**Figure 5: Improved detection after preprocessing and fixes**

Post-$\log_2$ filtering and NMS applied – clean, accurate bounding boxes.

- Example: Surgical_implant [1] and dental [3] detected with high confidence (e.g., 0.90+).
- Success: Proper localization without overlaps or artifacts → aligns with best results (F1 0.503 without $\log_2$, no augmentation – see poster Table 1). Demonstrates iterative improvement from feedback (progress form, sprint 4).

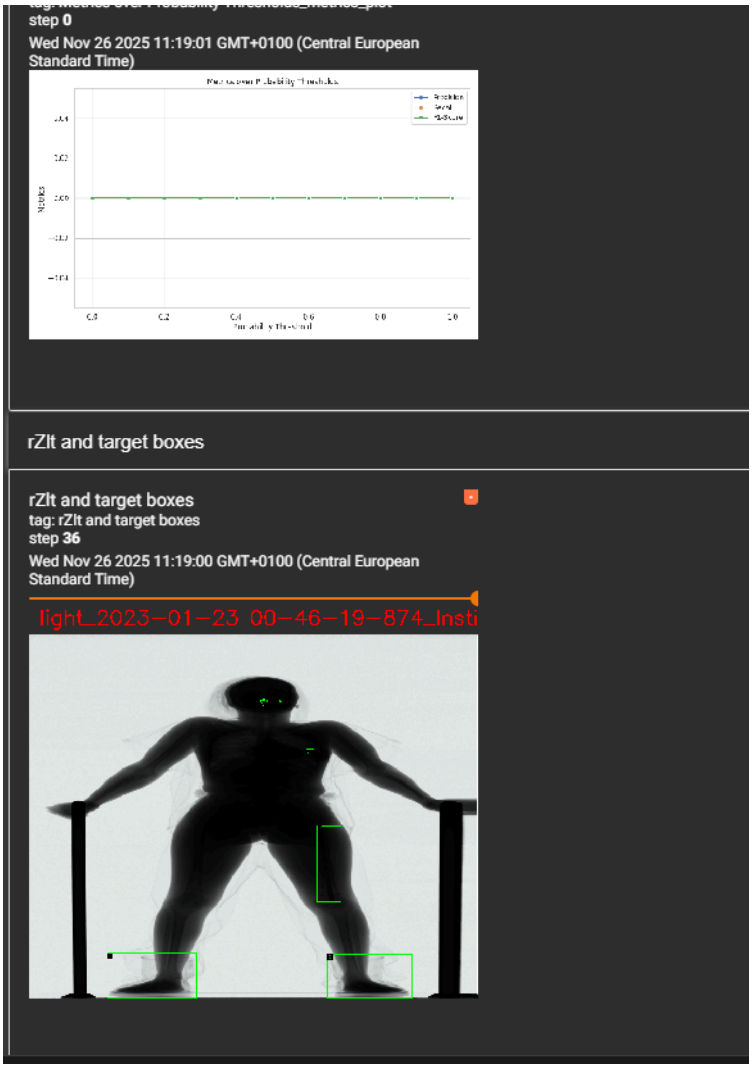


**Figure 6**: Output of a Raw image wuth ground truth  via Tensorboard.

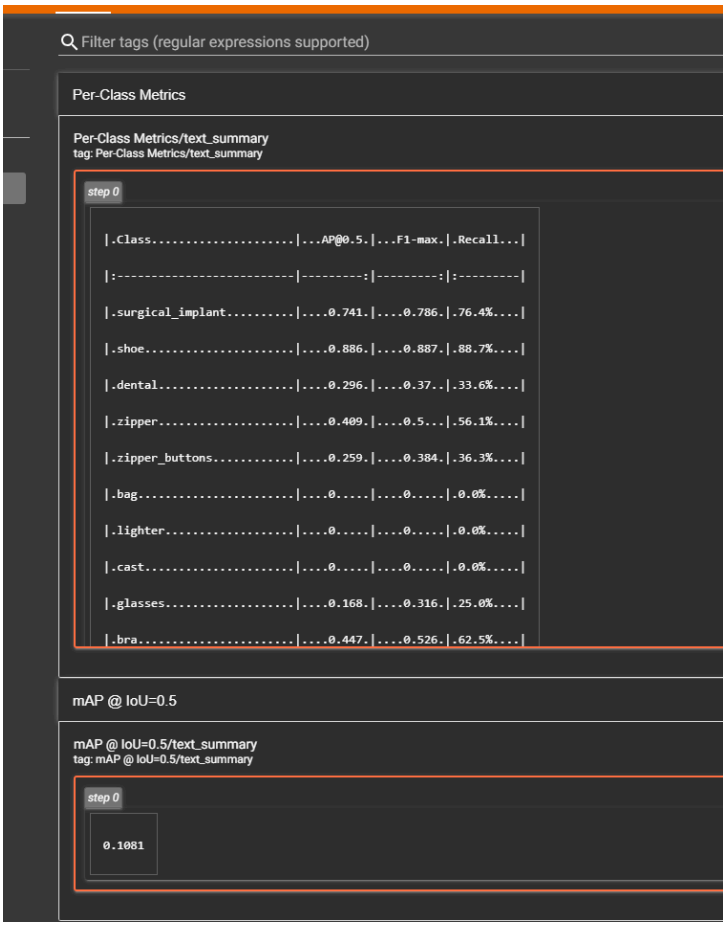

**Figure 7**: An image showing the matric per class results for each classes via TensorBoard.

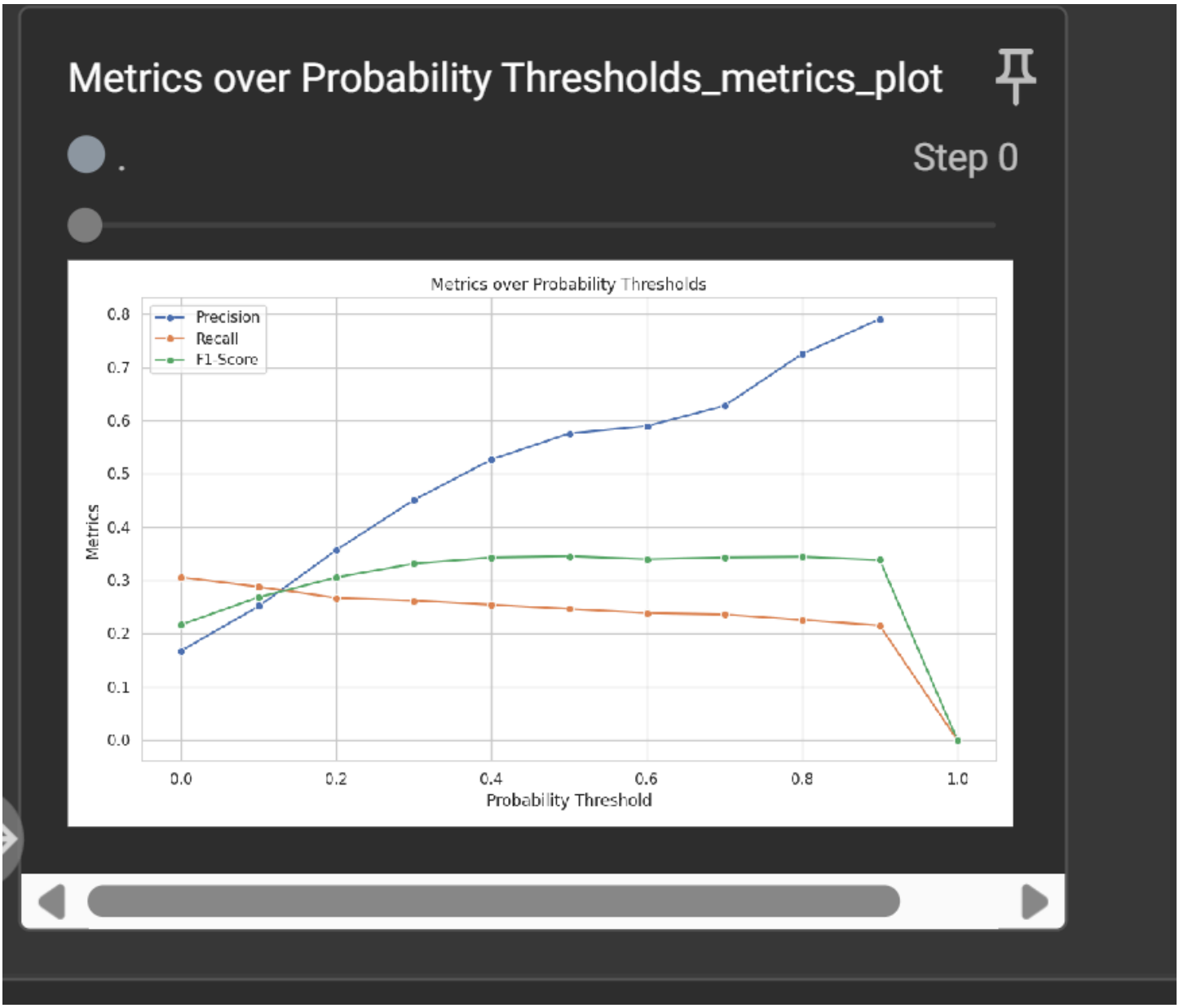


Figure 8: Map over Probability Threshold graph via TensorBoard.

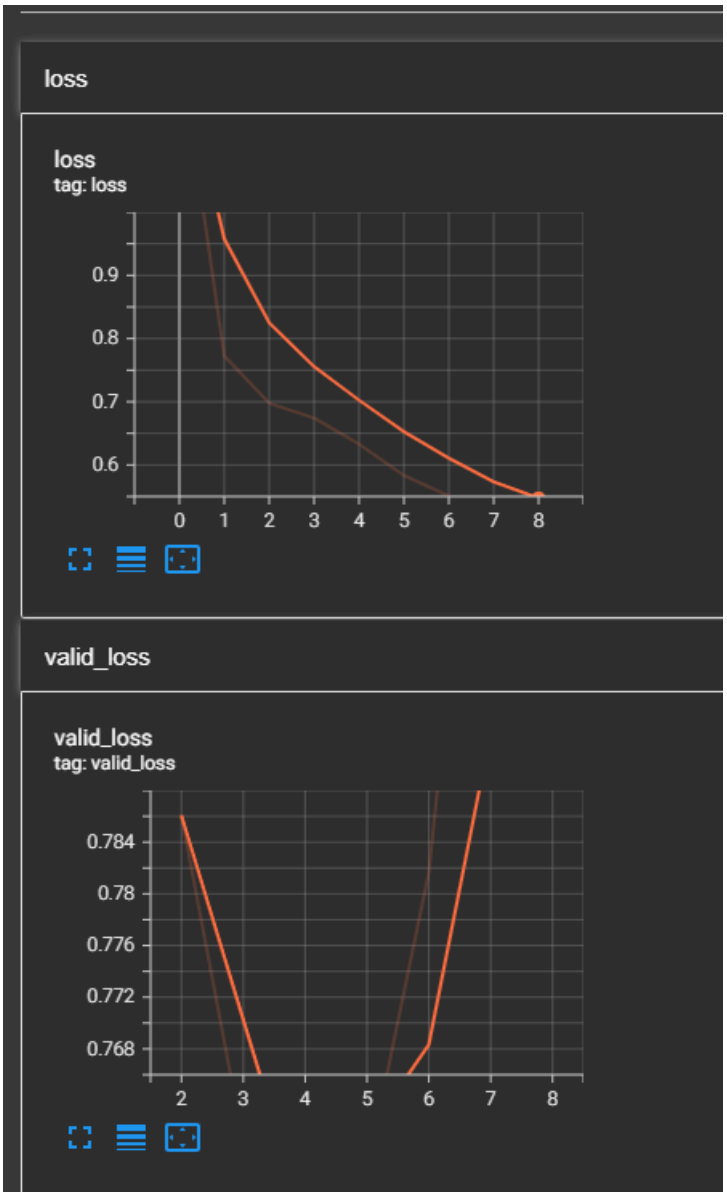


**Figure 9**: An Image the loss and validation loss during training via TensorBoard.